Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/06/2017
PI Name:	Hoffman, Jeffrey Ph.D.		
Project Title:	Quantifying and Preventing EVA Injury in Exploration Environments		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jhoffma1@mit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-452-2353
Organization Name:	Massachusetts Institute of Technology		
PI Address 1:	Aeronautics and Astronautics		
PI Address 2:	77 Massachusetts Avenue, 37-227		
PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
Start Date:	10/18/2016	**End Date:**	06/30/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Project ended early per PI--original end date was 10/17/2018 (Ed., 7/20/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Guillermo, Trotti M.A. (Trotti & Associates, Inc.) Anderson, Allison Ph.D. (University of Colorado, Boulder)		
Grant/Contract No.:	NNX17AB11G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR'S NOTE 7/23/2018: The Human Research Program (HRP) is delaying research on risks associated only with Mars surface operations or long term health, under which this project falls; thus, this project ended early in June 2017 (Information per M. Arya/JSC HRP) Gas-pressurized space suits cause injuries to astronauts both during training and actual extravehicular activities (EVAs) and significantly increase metabolic expenditure. Current astronaut injury protection systems are inadequate because they are 1) not easily customizable or adjustable, and 2) have not prevented injury thus far. As we utilize various new exploration and training environments, we will likely see new types of injuries accruing to astronauts unless improved protection systems are created. In our previous work we have evaluated astronaut injury and developed prototype protective devices to improve comfort and suit fit. We quantified pressure over the upper arm as a person moves inside the space suit, and revealed body-suit contact and suit articulation. We have identified the most relevant factors attributed to shoulder injury, including anthropometric dimensions, indicating individual suit fit is critical to preventing injuries. Our methods measure performance from a detailed look inside the suit (i.e., the perspective of the astronaut), rather than only analyzing suit motion externally. The product of our research will be to develop an Adjustable Protection Garment to improve astronaut comfort and decrease injury. To ensure the garment's performance, we will develop an Injury Monitoring System to measure human-suit contact and kinematics over the entire body. These systems will be tested in realistic EVA environments (i.e., in the neutral buoyancy lab, on the ARGOS facility, and/or in planetary exploration mobility tasks) to assess astronaut injury and discomfort. We will quantify pressure 'hot spots', assess comfort, and reduce injury potential during EVA tasks with and without the protection garment to ensure that the proposed wearable garment does not cause a performance decrement. Leveraging our previous research on injury analysis, musculoskeletal modeling of EVA injuries, countermeasure systems design and prototyping, we will develop second generation comfort and protection design concepts and a suit-contact sensing system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2017. EDITOR'S NOTE 7/23/2018: HRP is delaying research on risks associated only with Mars surface operations or long term health, under which this project falls; thus, this project ended early in June 2017 (Information per M. Arya/JSC HRP)
Bibliography Type:	Description: (Last Updated:)